

Indian Karting Race 2025

Season 10.0



Team Name: Team Engines and Demons

Team ID: IKR25-CV-2981

Report Title: Group-A Design Report

Report Subtitle: Go-Kart Engineering Design Report - IKR 2025

Abstract—This report presents the comprehensive design, analysis, and fabrication approach for a student-built go-kart, developed with the primary objectives of safety, performance, reliability, and cost-effectiveness. The design process involved detailed study and integration of various subsystems, including chassis, steering, braking, powertrain, and ergonomics. This go-kart reflects the application of engineering fundamentals, creative problem-solving, and team-based project execution. The report further highlights challenges encountered and design decisions made during the project lifecycle.

I. INTRODUCTION

The design and construction of a go-kart is a multifaceted engineering challenge that combines principles of mechanical design, material science, and vehicle dynamics. This project aimed to develop a functional and competitive go-kart, focusing on maximizing performance while ensuring safety, efficiency, and cost-effectiveness. The go-kart is designed for competitive purposes, with a keen focus on the optimal integration of various subsystems such as the chassis, powertrain, steering, suspension, and braking systems.

The design process was initiated with a thorough analysis of performance requirements, vehicle stability, and safety standards, followed by the conceptualization of the go-kart's core components. Emphasis was placed on creating a lightweight yet durable structure to enhance speed, maneuverability, and overall handling. In addition, the design team sought to balance the power output with fuel efficiency and ease of maintenance.

The project involved the collaborative efforts of a multidisciplinary team, utilizing tools such as Computer-Aided Design (CAD) software for modeling and Finite Element Analysis (FEA) for structural validation. This report documents the complete design process, detailing the methods employed, challenges faced, and solutions implemented to achieve a high-performing go-kart.

II. CHASSIS DESIGN REPORT

A. Rulebook Constraints

- Chassis material has to be either aluminium or steel and in case of steel, minimum carbon percentage has to be 0.1
- The chassis is to be constructed using seamless tubes with a circular cross section of wall thickness ranging from 1-2 mm with outer diameter ranging from 1-2 inches.
- The use of leaf springs, solid bars, I-bar, rectangular bar, or any other shape is strictly restricted.
- An open end in the chassis has to be kept in the frame which needs to be capped and also a 3 mm drill has to be provided on the roll hoop for thickness inspection.
- There has to be a ground clearance of maximum 1 inch (without driver inside) and 5 inches (with driver inside).
- The vehicle must have 2 jack points (front and rear side) and 2 hitch points (front and rear side) painted in yellow and orange respectively.
- The roll hoop must be made using seamless tube. Same chassis material with no wrinkles and no change in

cross section after bending which has to be supported by trusses/braces.

- Bumpers must be placed on front, rear and side such that covers the tire made up of seamless tubes of chassis material and attached to the frame by at least 4 points.

B. Dimension Parameters

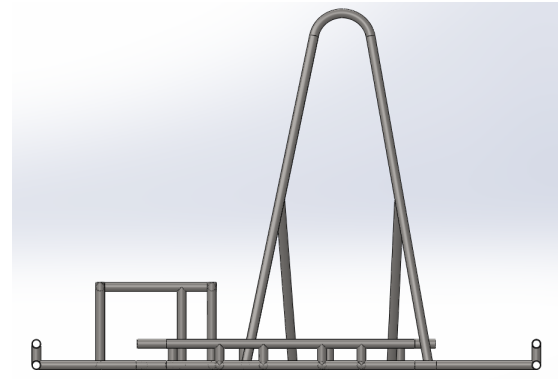


Fig. 1: Front View of Chassis

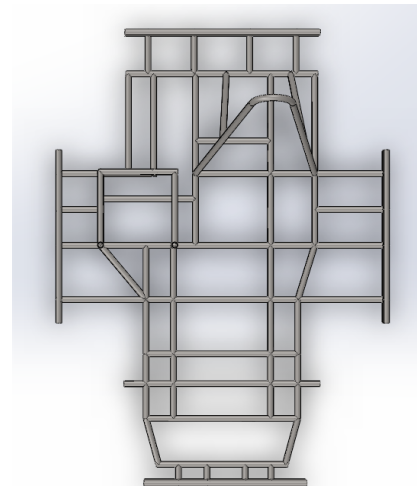


Fig. 2: Top View of Chassis

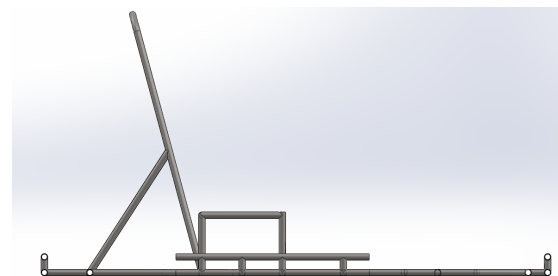


Fig. 3: Side View of Chassis

Following are the design Parameters:

- **Bounding Box Dimensions:**
- Length:2000

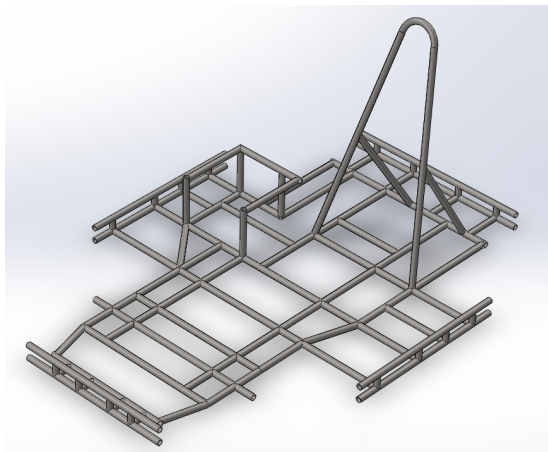


Fig. 4: Isometric View of chassis

- Width: 1500
- Height: 1100
- Wheelbase: 1200 mm
- Front Track Width: 1100 mm
- Rear Track Width: 1150 mm
- Ground Clearance: 3.75inch - 4inch
- Chassis Type: Hybrid
- Chassis Weight: 32-35 kg
- Total Go-Kart Weight with Driver: 170-190 kg
- Center of Gravity (CG): $X = 652.015678$, $Y = 60.235563$, $Z = -13.495907$

C. Design Considerations

• Rulebook Compliance

- The wheelbase we have chosen is greater than 40 inches (1016mm) and the smallest trackwidth is 1100 mm which is greater than 65 percent of the wheelbase which satisfies the rulebook constraint.
- The material AISI 1018 is chosen as chassis material which passes the material composition criterion.

• Chassis Geometry and Layout

- The wheelbase of the kart is 1200 mm which ensure a good balance between stability and maneuverability.
- The front track width of 1100 mm and rear track width of 1150 mm which is adequate enough for grip.

• Weight Distribution and Center of Gravity

- It was aimed to keep the front-to-rear weight distribution of 40:60 and hence used rear-engine setup.
- The engine mounts and seat mounts in the chassis have been placed in order to keep the center of gravity of the go-kart low.
- The lateral center of gravity should be as close to the centerline of the kart as possible.

• Structural Strength and Stiffness

- The chassis has been designed and the members have been added keeping in mind that it will sustain the impulsive, compressive, torsional and bending loads.

- The balance between flexibility and stiffness has also been maintained which was ensured by the selected material AISI 1018 carbon-steel.

• Safety Provisions and Mounting Interfaces

- An attempt has been made to integrate the mounts for mandatory safety components like bumpers, fire extinguisher, kill switch.
- The sharp edges and protrusions have been avoided in order to prevent any kind of injuries during operation or crashes and critical mounting areas are reinforced to prevent failure under impact or fatigue.

• Manufacturing and Assembly

- The manufacturing and fabrication of the chassis has also been kept in mind while designing the chassis so it makes easy for welding and design modular for simplifying the maintenance and replacement of any damaged parts.

• Simulation and Testing

- Finite Element Analysis and impact tests have been performed to evaluate stress, deformation and validate the structural integrity for the race-like loading conditions.

D. Material Selection

- The selection of material was done keeping in mind design constraints like the composition of material, final weight of chassis and dimension restraints and also keeping in mind the price limitations.

Materials	Yield Strength (MPa)	Percentage Elongation at Break	Cost per kg (Rs)
AISI 1026	260–440	17–27%	345
AISI 4130	435–979	18–26%	735
AISI 1020	230–370	18–28%	315
AISI 1018	270–400	18–29%	300

TABLE I: Material Properties and Cost

- The materials we considered were AISI 1018, AISI 4130, AISI 1026 and AISI 1020 among which AISI 1020 is not much durable at the price it is available while AISI 1018 has better strength at around same price range; AISI 4130 has high strength to weight ratio but due to high carbon percentage, it is more vulnerable to machining and at the same time it is very expensive.
- The modulus of elasticity of AISI 1018 is similar to that of AISI 4130 and at the same time, AISI 4130 is highly expensive so using AISI 1018 won't affect the weight and stiffness in a member of same geometry.
- The material we chose is AISI 1018 carbon steel. It has a good carbon content of 0.14-0.2 percentage which gives the material a sufficient hardness and yield strength, maintaining a good balance between ductility and toughness. It also has an excellent weldability and machinability and hence considered as best steel for carburizing parts.

E. Methodology

• Understanding Rulebook and Requirements

- The official IKR 2025 rulebook have been studied thoroughly to identify the dimensional, material, safety and all the design constraints and have proceeded accordingly.
- The key performnace parameters such as high torsional rigity, balanced weight distribution, safety of kart as well as the driver and ease of manufacturability are defined respectively.

• Conceptual Design

- Initial iteration of the chassis was made keeping in mind trackwidth, wheelbase and rough layout of major components
- After considering the factors such as driver ergonomics, weight balance and the mounting of the engine and the steering assembly, the next iterations created.
- The lateral center of mass has been kept as close to the centerline of the kart as possible to ensure symmetric handling behaviour.
- Weight balance in the longitudinal direction is aimed at 40:60 ratio to ensure adequate traction from the rear wheels.

• Material Selection

- AISI 1018 mild steel was selected as our chassis material due to its good availability, machinability, good mechanical strength and the cost effectiveness.
- The tube with outer diameter of 1 inch and wall thickness of 2 mm was been chosen for construction of the roll cage which came out as the near correct tube dimensions optimizing the strength-to-weight ratio.

• 3D CAD Modeling

- Solidworks have been used as the CAD software for modelling the chassis, adhering to all dimensional constraints and safety provisions specified by the Indian Karting Race 2025
- Custom structural members have been created to create the model of the chassis after creating 3D sketches.
- Chassis was edited multiple times according to analysis results. Weight was kept in mind while adding or removing key members.

• Structural Member Orientation

- Orientation or structural members determines load transfer and structural integrity in the 2 axes of the kart
- Lateral members were kept intact and were joined by longitudinal members which were cut to their respective lengths
- This ensures safety in lateral crash cases and efficient load distribution of main loads i.e driver and engine

• Finite Element Analysis (FEA)

- Structural analysis was performed using ANSYS Mechanical to evaluate:

- Static loading conditions (driver weight, cornering, braking, etc.)
- Dynamic loading conditions (crash situation)
- Torsional rigidity of the chassis under diagonal loading
- Maximum stress locations and corresponding safety factor

• Optimization

- Based on the FEA results, suitable iterative design improvements were made to get an optimized design with strengthened critical areas and reduced unnecessary mass.

• Validation

The final design was evaluated for the following parameters:

- Maintaining the torsional stiffness within the acceptable range.
- Rulebook compliance has been kept in mind for all the geometry and safety requirements.

• Documentation and Manufacturing Planning

- Detailed engineering drawings were prepared for fabrication.
- Jigs and fixtures were required for manufacturing particular parts and it was planned accordingly.
- The welding sequences and fabrication procedures were outlined respectively for quality control.
- Tack welds were initially done to ensure dimensional accuracy of the chassis, followed by seam welds. Welding performed was MIG welding.

F. Roll Cage Analysis

- A dynamic crash analysis was performed on the chassis structure to evaluate the strength and safety of the roll cage under a frontal impact scenario
- The simulation modeled the vehicle striking into the wall with a specified velocity, replicating crash conditions for impact testing
- Front and rear impacts were simulated 30kmph and side impacts were simulated at 10kmph keeping in mind realistic scenarios

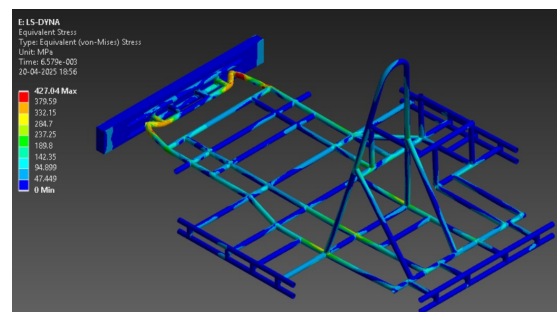


Fig. 5: Dynamic Crash Test of Chassis(Front)

G. Results

- Max Equivalent (Von Mises Stress) is found to be 427.04 MPa.

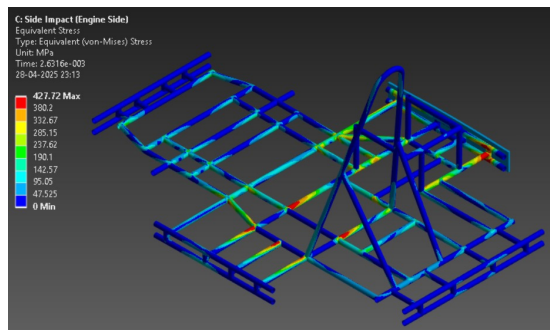


Fig. 6: Dynamic Crash Test of Chassis(Side)

- Closer analysis of the results shows that the front section (ahead of the pedals and foot guard) is absorbing majority of the energy, and may yield as a result. This ensures that the driver is safe.
- Stress paths show the transfer of stresses throughout members of the chassis. Longitudinal members distribute the front and rear impact energy, whereas the lateral members distribute side impact forces.
- In both cases, load is kept away from the roll hoop and driver cell, to ensure maximum safety

III. STEERING DESIGN REPORT

A. Introduction

The steering design is essential in providing responsiveness, control and a good driver handling experience. The steering report includes the basic geometry that went into designing the Ackermann geometry as well as a detailed Finite Element study of the knuckle under bump and braking conditions along with Driver effort calculations.

B. Rulebook Constraints

The specified rulebook constraints include:

- The steering wheel must be mechanically connected to the wheels.
- All parts must be made of metal except for the steering wheel.
- Steering system must have positive steering locks.
- The maximum turning radius is given to be 3 m.
- Minimum wheelbase of 1016 mm.
- The smaller trackwidth must be atleast 65 percent of the wheelbase.

C. Dimension Parameters

The dimensions of the Kart are:

- A wheelbase of 1200 mm
- A front trackwidth of 1100 mm
- A rear trackwidth of 1150 mm

The above dimensions were chosen in order to provide a smoother handling experience as well as keeping in mind comfort of the driver.

Besides this, a camber angle of 0° and a castor angle of 15° has been chosen, which can be changed based on feedback obtained during test runs.

D. Steering Geometry

- The art uses an Ackermann steering geometry for tighter turning since it is for a lower speed application.
- While turning the inner wheel has to follow a circle with a smaller radius, the outer wheel has to follow a circle with a larger radius.
- So, Ackermann steering geometry allows different turning angles at different wheels.

E. Design Considerations

- In the Ackermann geometry, a 2:1 steering ratio has been given, which is adjustable to 1:1.
- A steering angle of 120° (lock to lock) is provided at the steering wheel
- A castor angle of 15° was provided to have a good steering feel and a good amount of self-centering torque which will be explained with the help of calculations.

F. Component Selection

Components of Steering Systems are as follows:

- Steering knuckle- Machined from EN-8 low-carbon steel. Design considerations like bump force, Brake force, and steering force were kept in mind while designing.

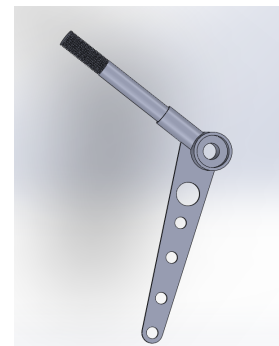


Fig. 7: Steering Knuckle top view

- Steering Wheel- The Steering Wheel was bought OTS, keeping in mind the rulebook constraints.
- Rod end ball joints- Rod end ball joints were used at the end of tie rods to maintain degrees of freedom of the tie rods.
- Pitman arm - The Pitman arm was designed and manufactured to transfer steering rotation to the wheels through tie-rods while maintaining the desired steering ratio.

G. Working Principle

The go-kart steering system uses a mechanical linkage to transmit the driver's input from the steering wheel to the front wheels.

- When the driver rotates the steering wheel, the motion is transferred through a steering column to the pitman arm.
- The pitman arm is connected to the Steering knuckle attached to the front wheels. So the motion of the pitman arm is then transferred to the Steering knuckle through tie rods, which results in the turning of the wheels.

- The system follows the principle of Ackermann steering geometry, ensuring:
 - During a turn, the inner wheel turns at a sharper angle than the outer wheel.
 - This minimises tire slippage and maintains correct wheel alignment with the turn radius.
- The geometry was achieved by designing the steering arm so that lines drawn through the steering axis and point of attachment of tie rods on the steering arm intersect at the centre of the Rear axle.
- Further, the steering system incorporates a positive caster angle, where the steering axis tilts rearward from vertical.
- The caster angle provides self-aligning torque, helping the steering wheel return to center after a turn and enhancing straight line stability and improving steering feedback provided to the Driver.

H. Formulation and Calculations

• Wheel Angles:

The following values are calculated for the steering geometry based on the Ackermann geometry. The steering angles for the inner and outer wheels based on Ackermann geometry are given by:

$$\tan(\theta_{in}) = \frac{L}{R - \frac{W}{2}} \quad (1)$$

$$\tan(\theta_{out}) = \frac{L}{R + \frac{W}{2}} \quad (2)$$

where:

- L is the wheelbase
- W is the trackwidth
- R is the turning radius from the center of the rear axle to the Center of rotation

The effective distance from the Center of rotation to the wheel contact point, denoted by e , is calculated as:

$$e = R_{mean} \cdot \cos\left(\frac{\theta_{in} + \theta_{out}}{2}\right) \quad (3)$$

• Force Calculations:

Assumptions:

Weight of the kart (With Driver) = **180 kg**

Weight Distribution = **40:60**

– Knuckle forces

* Bump Force:

$$F_{BUMP} = W_{KART} \times 0.2 \times 3g \quad (4)$$

Here:

An assumption is taken into account that a bump force of $3g$ is applied to the wheel, where g is the acceleration due to the gravitational force. Therefore,

$$F_{BUMP} = 1059.48N$$

* Braking force:

$$F_{BRAKE} = W_{KART} \times 0.2 \times g \quad (5)$$

Here:

Deceleration due to braking is assumed to be $1g$, where g is the acceleration due to gravity. Therefore,

$$F_{BRAKE} = 353.16N$$

• Driver Effort and Self-aligning Torque at Wheel

– Self aligning torque at wheel

* Mechanical trail:

$$M_T = R_W \times \tan(Ca) \quad (6)$$

Where:

- M_T is the mechanical trail
- R_W is the Radius of the front wheel
- Ca is the caster angle

* Self-aligning torque:

$$F_o = \frac{W_{KART} \times 0.2 \times v^2}{r} \quad (7)$$

Where:

- F_o is the Frictional force on the front wheel
- v is the velocity of the Go-kart
- r is the turning radius

$$M_o = F_o \times M_T \times \tan(Ca) \times \cos(\gamma) \quad (8)$$

Where:

- M_o is the self aligning torque
- M_T is the mechanical trail
- Ca is the caster angle
- γ is the wheel angle

– Driver Effort

$$F_{driver} = \frac{M_{oL} + M_{oR}}{2 \times R_{sw}} \quad (9)$$

Where:

- F_{driver} is the force applied by the driver on the steering wheel
- M_{oL} is the Self aligning torque due to left wheel
- M_{oR} is the Self aligning torque due to right wheel
- R_{sw} is the radius of the steering wheel

I. Results

• Turning radius and Wheel angles

Parameter	Value
Wheelbase (mm)	1200
Front Trackwidth (mm)	1100
Rear Trackwidth (mm)	1150
R_{mean} (mm)	2500
Offset distance from WC (mm)	130
e (mm)	1618
R_i (mm)	2015
R_o (mm)	3017
Inner Wheel Angle (°)	36.56
Outer Wheel Angle (°)	23.44
Steering Ratio	2
Steering wheel angle(°)	60

TABLE II: Ackermann Steering Geometry Parameters

• Driver steering effort

Turning Radius (m)	Speed (km/h)	Driver Effort (N)
2.5	10	25.47
2.75	15	53.90
3	20	89.93
5	25	90.21
6.625	30	99.32

TABLE III: Driver steering effort for different speeds and turning radii

• Steering knuckle strength validation

– Under bump conditions

Parameter	Value
Force applied	1059.48 N
Max. Stress developed	255.64 MPa
Max. Deformation	0.47 mm
Min. Factor of Safety	1.1481

TABLE IV: Knuckle Structural analysis parameters (Bump)

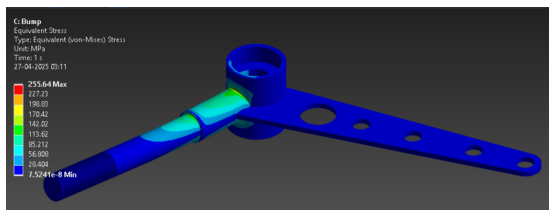


Fig. 8: Stress contour under bump conditions

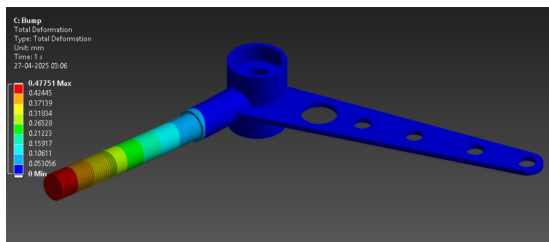


Fig. 9: Deformation contour under bump conditions

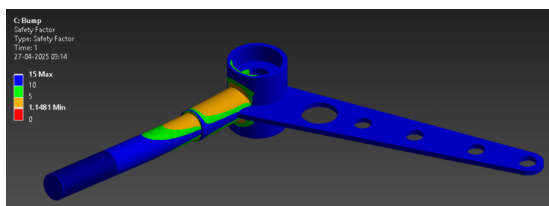


Fig. 10: Factor of safety contour under bump conditions

– Under braking conditions

Parameter	Value
Force applied	353.16 N
Max. Stress developed	95.153 MPa
Max. Deformation	0.16 mm
Min. Factor of Safety	2.64

TABLE V: Knuckle Structural analysis parameters (Braking)

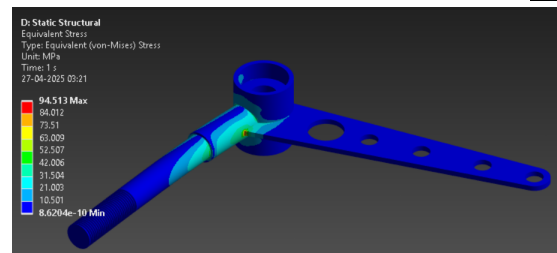


Fig. 11: stress contour under braking conditions

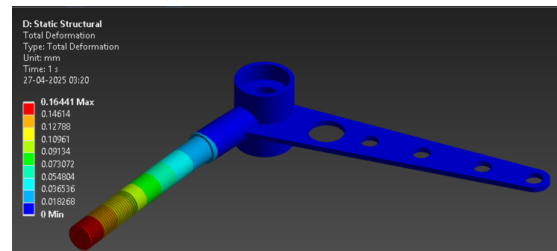


Fig. 12: Deformation contour under braking conditions

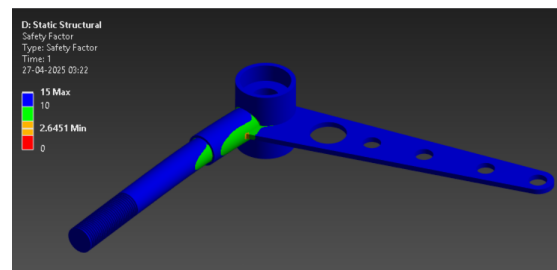


Fig. 13: Factor of safety contour under braking conditions

IV. BRAKING SYSTEM DESIGN REPORT

A. Introduction

Braking is one of the most essential subsystems in any vehicle. For the kart, a braking system is designed such that it adheres to all rulebook guidelines and ensures driver safety at all times. The system used in the go-kart is a pedal-actuated hydraulic braking system. Various design constraints and analysis results were considered during the development process, the details of which are outlined in the report below.

B. Rulebook constraints

1) Brake system constraints:

- The system must be hydraulically controlled only. Brake by wire is strictly prohibited.
- The brakes should act on both rear-wheels at the same time.
- Use of a brake fluid reservoir is compulsory. Use of steel braided brake hose and brake lines in compulsory.

2) Brake pedal constraints:

- Pedal should be fabricated only from steel, aluminum or titanium.
- Pedal should be foot operated only.
- Brake pedal must withstand a force of 2000N. Also its motion should be restricted by using some kind of pedal locks.

3) Technical Inspection constraints:

- Both rear wheels must lock on the application of the brakes.
- During the test, the speed of the vehicle must be maintained at 25kmph minimum.
- Test Procedure - The driver drives in a straight line for 50m and then applies the brakes as soon as he crosses the line. The brake should be applied completely and after application the vehicle should travel in a straight line only.

C. Force considerations

The following forces and their effects were considered during calculations:

- Maximum deceleration caused on complete application of the brakes.
- Weight distribution on the front and rear wheels during static loading.
- Frictional forces acting between road and wheel, and between brake pad and brake disc.

D. Working principle

The braking system operates on the principle of hydraulic pressure transfer. When the driver presses the brake pedal:

- The Force in pedal is transmitted to the master cylinder with a ratio of 1:6, which means force is scaled 6 times.
- The master cylinder converts this mechanical force into hydraulic pressure.
- Hydraulic fluid transfers pressure to the caliper.
- Caliper pistons press the brake pads against the rotating disc.

E. Design Considerations

Following factors were considered:

- Efficient heat dissipation from the brake disc.
- Minimal pedal effort for maximum braking efficiency.
- Easy maintenance and part replacement.
- Compactness and weight optimization.
- Mounting and alignment with axle.

F. Component Selection

The components of the Go-Kart brake system are as follows:

- Brake Pedal - Machined from EN-8 low carbon steel. Design considerations like maximum pedal force of 2000N and pedal ratio were kept in mind while designing.
- Master Cylinder, Fluid reservoir, Brake Calipers and Brake pads - Acquired from a Go-Kart manufacturer (Vintage-Vision, Indore).
- Brake Lines - Metal, steel braided brake fluid hoses used for safe and effective usage of the brake fluid.
- Brake Fluid - Standard brake fluid (DOT-4).
- Brake disc rotor - Designed considering the dimensions of the brake caliper and incorporating vents/holes for effective heat dissipation. Material used was SS410.
- Rotor Hub - This part is used to mount the brake disc rotor onto the driveshaft. This makes sure that both rear wheels lock simultaneously on the application of the brakes. Material used is AL6061-T6.

G. Pictures of design components and assembly

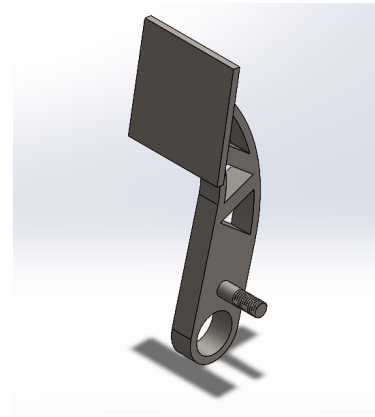


Fig. 14: Brake Pedal Isometric View



Fig. 15: Brake Disc Front View

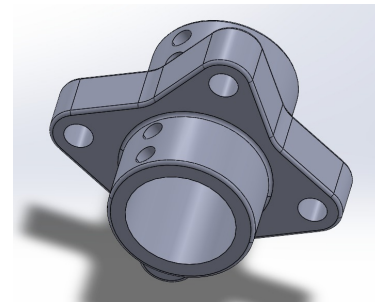


Fig. 16: Hub for Disc

H. Formulation and Calculations

The following calculations were made during the brake system design:

- Pedal Ratio:

$$\eta = \frac{\text{Input pedal arm length}}{\text{Output pedal arm length}} = \frac{1}{6}$$

- Brake force calculation on the brake disc:

$$F_{\text{brake}} = \frac{F A_{cp}}{\eta A_{mc}} \approx 4882.5$$

Where,

$$F_{brake} = \text{Normal force at disc}$$

$$F = \text{Force applied by driver (250N)}$$

$$A_{cp} = \text{Area of caliper piston (962.1 mm}^2\text{)}$$

$$A_{mc} = \text{Area of master cylinder (295.6 mm}^2\text{)}$$

- Decelerating force:

$$F_d = 2\mu \frac{rF_{brake}}{R} = 1817.4N$$

Where,

$$\mu = \text{Frictional coefficient between pad and disc (0.4)}$$

$$r = \text{Point of application of } F_b \text{ (65mm)}$$

$$R = \text{Radius of rear wheel (139.7mm)}$$

- Deceleration caused by the brakes:

$$a = \frac{F_d}{m} = 10.1 \text{ m/s}^2$$

- Stopping Distance (assuming $v = 60 \text{ km/hr}$):

$$\frac{v^2}{2a} = 13.75 \text{ m}$$

- Heat flux calculation (used in the thermal analysis of the brake disc):

- 1) Total kinetic energy of the vehicle before the application of the brakes:

$$KE = \frac{1}{2}mv^2$$

- 2) This kinetic energy is then assumed to be completely converted to heat which is then given to the brake disc. The heat flow is obtained as:

$$Q_{flux} = \frac{1}{2A_p}mv^2$$

Where,

$$Q_{flux} = \text{Heat Flux}$$

$$A_p = \text{Area of brake pad}$$

I. Part Analysis

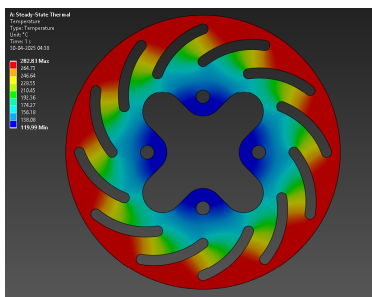


Fig. 17: Steady state thermal analysis of the Brake Disc

Parameter	Value
Heat Flux (on both faces)	29000 W/m ²
Convection film coefficient	75 W/m ²
Ambient Temperature	30 C°
Max. Temperature	282.83 C°
Min. Temperature	119.99 C°

TABLE VI: Parameters used for steady state thermal analysis of brake disc

J. Results

- Master Cylinder Force = 1500 N
- Deceleration = 10.1 m/s²
- Stopping Distance = 13.75 m

V. TRANSMISSION DESIGN REPORT

A. Introduction

The transmission system is responsible for transferring power from the engine to the wheels efficiently. A transmission that ensures smooth power delivery, good acceleration, and reliability during the race has been engineered. The design focuses on simplicity, performance, and ease of maintenance while following all event rules. This section covers our transmission layout, sprocket ratio selection, and overall design approach.

B. Rulebook Constraints

The following constraints are specified under Section C.4.1 of the IKR 2025 Rulebook:

1) Drive Type (Rule C.4.1.1.1):

The transmission system must be strictly of **rear-wheel drive** only.

2) Allowed Transmission Types (Rule C.4.1.1.2):

The teams are free to use any type of transmission system – gear box/ CVT/ Chain and sprocket attached to a centrifugal clutch.

3) Power Delivery Methods (Rule C.4.1.1.3):

Power may be transmitted to the rear wheels using:

- Differential
- Through axle
- Wheel mounting hub
- Any other structurally safe and feasible method

4) Sprocket Ratio (Rules C.4.1.1.4 & C.4.1.1.5):

- There is **no restriction** on the sprocket ratio.
- Teams are allowed to **modify sprocket ratios** as needed.
- Teams must submit a **calculation report** supporting the selection of the sprocket ratio.

C. Force considerations

The following calculations are based on the TVS Fiero FX 150 engine with the given specifications:

- Maximum Power: 12 HP @ 7000 RPM
- Maximum Torque: 10.5 Nm @ 6500 RPM
- Engine Displacement: 147.5 cc
- Transmission: 4-speed gearbox

1) Assumed Transmission Setup:

- Driver Sprocket (Engine): 13 teeth
- Driven Sprocket (Axle): 39 teeth
- Chain Pitch: 15.875 mm (Standard 520 chain)
- Final Reduction Ratio: 3:1

2) *Pitch Radius of Driver Sprocket:* The pitch radius r is calculated using the formula:

$$r = \frac{P}{2 \cdot \sin\left(\frac{3.1416}{N}\right)}$$

where:

- $P = 0.015875$ m
- $N = 13$

$$r = \frac{0.015875}{2 \cdot \sin\left(\frac{3.1416}{13}\right)} \approx 0.03317 \text{ m}$$

3) *Pitch Radius of Driven Sprocket:* The pitch radius r is calculated using the formula:

$$r = \frac{P}{2 \cdot \sin\left(\frac{3.1416}{N}\right)}$$

where:

- $P = 0.015875$ m
- $N = 39$

$$r = \frac{0.015875}{2 \cdot \sin\left(\frac{3.1416}{39}\right)} \approx 0.09864 \text{ m}$$

4) Tangential Force on Chain (Tight Side):

$$F_{tight} = \frac{T}{r} = \frac{10.5}{0.03317} \approx 316.55 \text{ N}$$

5) *Slack Side Tension:* Assuming slack side tension is 20% of the tight side:

$$F_{slack} = 0.2 \times 316.55 = 63.31 \text{ N}$$

6) Total Chain Tension:

$$F_{total} = F_{tight} + F_{slack} = 316.55 + 63.31 = 379.86 \text{ N}$$

7) *Chain Speed at 7000 RPM:* Convert RPM to angular speed:

$$\omega = \frac{2\pi \times 7000}{60} \approx 733.04 \text{ rad/s}$$

$$v = \omega \cdot r = 733.04 \times 0.03317 \approx 24.31 \text{ m/s}$$

8) Power Transmitted:

$$P = F_t \cdot v = 316.55 \times 24.31 \approx 7695.33 \text{ W} \approx 10.32 \text{ HP}$$

Parameter	Value
Engine Torque	10.5 Nm
Tangential Force on Chain	316.55 N
Slack Side Tension	63.31 N
Total Chain Tension	379.86 N
Chain Linear Speed	24.31 m/s
Power Transmitted	10.32 HP
Reduction Ratio	3:1

TABLE VII: Summary of Chain-Sprocket Transmission Forces

D. Working Principle

The transmission system operates on the working principle of mechanical power transfer through rotational motion. When the engine is running and the throttle is applied:

- The torque (rotational power) is produced by the engine at the crankshaft by burning fuel.
- The torque is passed directly to a drive sprocket on the engine's output shaft.
- This rotational motion is imparted to the driven sprocket fixed on the rear axle by a roller chain.
- The ratio of the drive and driven sprockets' gears will determine the speed and torque transmitted to the wheels.
- The rear axle turns, making the back wheels rotate and moving the go-kart ahead.

E. Design Considerations

Designing the transmission system was done keeping in mind performance, reliability, and safety. The following most important factors were taken into account during the designing process:

- A chain and sprocket system was chosen due to its simplicity, light weight, and adaptability for go-kart operations.
- The system does not incorporate a centrifugal clutch, and hence it allows direct power transfer from the engine to the rear axle.
- Gear ratio was specifically chosen with respect to track configuration to meet both acceleration and maximum speed requirements.
- Sprocket alignment was maintained to avoid friction losses and chain derailment during running.
- The tension of chain was selected such that it achieved to ensure smooth transmission and reduce wear on the parts.
- Wear-resistant materials were used for chains and sprockets to survive racing conditions and provide reliability.
- The design provides easy access for quick maintenance and adjustments under competitive conditions.
- All rotating and moving components are shielded to ensure driver safety and meet IKR 2025 regulations.

F. Component Selection

The transmission system is designed to ensure optimal power delivery from the TVS Fiero FX 150 engine to the rear wheels while maintaining simplicity, efficiency, and ease of maintenance. The components were selected based on engine output characteristics, race performance requirements, and rulebook constraints.

1) Engine:

- Single stroke - 147.5 cc - petrol engine
- **Maximum power:** 12HP @ 7000 RPM
- **Maximum torque:** 10.5Nm @ 6500 RPM
- 4 speed gearbox

2) *Sprockets:* After calculations for the optimal sprocket ratio, i.e., 3:1 (39:13), the stock sprocket set of TVS Fiero FX 150 was selected which provides the same number of teeth as required for our transmission.

- **Driver Sprocket (13 Teeth):** Selected to match the engine output shaft and provide high durability under high-speed conditions.
- **Driven Sprocket (39 Teeth):** Chosen to give a final reduction ratio of 3:1, balancing torque multiplication and top speed. This ratio ensures better low-end torque for quick acceleration on track.

3) Chain:

- **Chain Type:** 520 standard roller chain.
- **Justification:** Widely available and used for all bike transmissions, strong enough to handle the calculated chain tension of approximately 379.86 N, and suitable for the 10.33 HP power range. Provides a good balance between strength and weight.

4) Clutch:

- **Type:** Integrated with 4-speed gearbox.
- **Justification:** Utilized the engine's stock wet multi-plate clutch system, which ensures smooth gear shifting, reliable engagement, and minimal power loss.

5) Axle and Hub:

- **Axle:** Solid 30 mm shaft selected for strength and resistance to torsional failure.
- **Hub:** Lightweight aluminum hubs (Al6061 T6) used to minimize rotating mass and improve acceleration.

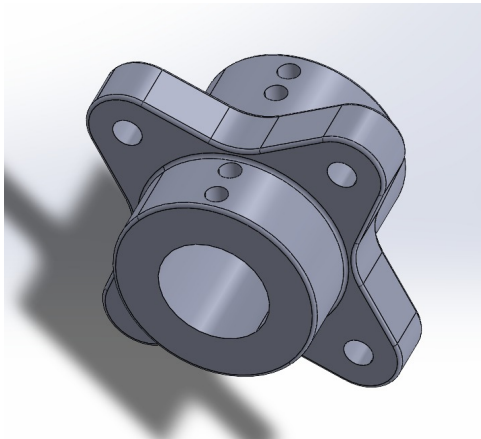


Fig. 18: Hub for driven sprocket

G. Dynamics Analysis / Simulation

The dynamic performance of the gearbox and transmission system is evaluated under various operating conditions to ensure optimized power delivery, efficient acceleration, and mechanical reliability.

1) Input Parameters

Parameter	Value
Mass of Go-kart (with driver)	180 kg
Engine Power (Max)	12 HP = 8948.4 W
Wheel Radius	0.139 m
Weight Distribution (F:R)	40:60
Coefficient of Friction (μ)	0.8
Rolling Resistance Coefficient (C_r)	0.005
Projected area	0.4 m ²
Drag coefficient	0.8
Drivetrain Efficiency	85%

TABLE VIII: Assumed parameters for tractive force and acceleration analysis

2) Maximum available traction on Rear Wheels

$$F_{\max} = \mu \times m_r \times g = 1 \times (0.6 \times 180) \times 9.81 = 1059.48 \text{ N}$$

3) Power Equation

Considering drivetrain efficiency ($\eta = 0.86$), available power:

$$P = 8948.4 \times 0.85 = 7606.14 \text{ W}$$

$$F_{\text{Resisting}} = F_{\text{rolling}} + F_{\text{air}}$$

$$F_{\text{rolling}} = 0.005 \times 180 \times g$$

$$F_{\text{air}} = \frac{1}{2} \times C_d \times A \times v^2$$

$$F_{\text{rolling}} = 8.829 \text{ N}$$

$$F_{\text{air}} = 0.16 \cdot v^2$$

$$P = F_{\text{Resisting}} \cdot v = (8.829 + 0.16 \cdot v^2)v$$

$$7606.14 = 8.829 \cdot v + 0.16 \cdot v^3$$

4) Maximum speed

$$7606.14 = 8.829 \cdot v_{\max} + 0.16 \cdot v_{\max}^3$$

Solving numerically, we get:

$$v_{\max} = 35.71 \text{ m/s} \approx 128.556 \text{ km/h}$$

5) Theoretical Acceleration Estimate at Low Speed

Parameter	Value
Primary Gear Ratio	3.25
Secondary Gear Ratio	3
Engine Torque (N·m)	10.5
Engine RPM	7000
Tire Radius (m)	0.1397
Coefficient of Rolling Resistance	0.005
Mass of Car (kg)	180
Frontal Area (m ²)	0.4
Rolling Resistance (N)	8.829
Drag Coefficient (Cd)	0.8
Air Density (kg/m ³)	1.204
Wind Speed (km/h)	23
Gear 1 Metrics	
Individual Gear Ratio	3
Overall Gear Ratio	29.25
Torque (N·m)	291.77
RPS	3.99
Speed (km/h)	5
Traction Force (N)	2088.54
Aerodynamic Drag (N)	0.31
Acceleration (m/s ²)	11.55

TABLE IX: Theoretically Possible acceleration as per engine metrics

6) Heat Generation and Dissipation

Assuming 5% of transmitted power converts to heat:

$$P_{\text{heat}} = 0.05 \times 7606.14 \text{ W} \approx 380.3 \text{ W}$$

Heat is dissipated through air cooling and conductive heat transfer to surrounding structures. No additional cooling systems are necessary.

7) Dynamic Load Analysis (Transient Conditions)

Under sudden load changes, peak torque on the chain can exceed steady-state values by approximately 10–15%.

Thus:

$$\text{Max Transient Tension} = 1.15 \times 379.86 \approx 436.84 \text{ N}$$

The selected 520 chain (rated for $\sim 1000 \text{ N}$) ensures safe operation under dynamic loads.

Parameter	Value
Maximum Axle Torque	31.5 Nm
Maximum Axle RPM	2333 RPM
Max Tractive Force	635.69 N
Power Delivered to Wheels	7606.14 W
Acceleration estimate at low speed ($v=5\text{m/s}$)	8.56 m/s ²
Max speed estimate	102.78 km/h
Maximum Chain Tension (Steady)	379.86 N
Maximum Chain Tension (Transient)	436.84 N
Heat Generation	380.3 W

TABLE X: Summary of Dynamic Performance Metrics

8) Conclusion

The gearbox and transmission system are designed to provide:

- High initial acceleration.
- Sufficient top-end speed.
- Durable, efficient, and low-maintenance operation.
- Safe operation under steady-state and transient load-ing conditions.

This dynamic analysis validates the system's capability to deliver strong, reliable, and race-ready performance.

VI. FORMULATION AND CALCULATIONS

The transmission system is analyzed to evaluate chain tension, power transmission efficiency, and vehicle acceleration. The following formulas were used:

A. 1. Pitch Radius of Sprockets

The pitch radius r is given by:

$$r = \frac{P}{2 \cdot \sin\left(\frac{\pi}{N}\right)}$$

where:

- $P = 15.875 \text{ mm}$ (chain pitch)
- $N = 13$ (driver), $N = 39$ (driven)

$$r_{\text{driver}} = 0.03317 \text{ m}, \quad r_{\text{driven}} = 0.09864 \text{ m}$$

B. 2. Chain Forces

$$F_{\text{tight}} = \frac{T}{r} = \frac{10.5}{0.03317} = 316.55 \text{ N}$$

$$F_{\text{slack}} = 0.2 \times F_{\text{tight}} = 63.31 \text{ N}$$

$$F_{\text{total}} = F_{\text{tight}} + F_{\text{slack}} = 379.86 \text{ N}$$

C. 3. Chain Speed and Power Transmission

$$\omega = \frac{2\pi \times 7000}{60} = 733.04 \text{ rad/s}$$

$$v = \omega \cdot r = 733.04 \cdot 0.03317 = 24.31 \text{ m/s}$$

$$P = F_t \cdot v = 316.55 \cdot 24.31 = 7695.33 \text{ W} \approx 10.32 \text{ HP}$$

D. 4. Tractive Effort and Acceleration

Assuming flat ground and no slip:

$$F_{\text{max}} = \mu \cdot m_r \cdot g = 0.6 \cdot (0.6 \cdot 180) \cdot 9.81 = 635.69 \text{ N}$$

Total tractive effort required:

$$F_{\text{total}} = ma + C_r mg + \frac{1}{2} \rho A C_d v^2$$

$$F_{\text{total}} = 180a + 26.49 + 0.29424v^2$$

E. 5. Power Equation

$$P = F_{\text{total}} \cdot v = (180a + 26.49 + 0.29424v^2)v$$

With effective power available:

$$P = 8948.4 \cdot 0.85 = 7606.14 \text{ W}$$

Solving:

$$\text{At } a = 0 \Rightarrow v_{\text{max}} \approx 28.55 \text{ m/s} (\approx 102.78 \text{ km/h})$$

$$\text{At } v = 5 \text{ m/s} \Rightarrow a = \frac{7436.91}{900} = 8.26 \text{ m/s}^2$$

F. 6. Dynamic Load Estimate

Assuming 15% increase in tension under dynamic loads:

$$F_{\text{transient}} = 1.15 \cdot F_{\text{total}} = 1.15 \cdot 379.86 = 436.84 \text{ N}$$

VII. RESULTS

The transmission system was evaluated based on static and dynamic performance metrics. The following results were obtained through theoretical calculations and simulations:

A. Chain-Sprocket Transmission Output

Parameter	Value
Engine Torque	10.5 Nm
Tangential Force on Chain	316.55 N
Slack Side Tension (20% of Tight Side)	63.31 N
Total Chain Tension (Steady State)	379.86 N
Chain Linear Speed @ 7000 RPM	24.31 m/s
Power Transmitted	10.32 HP
Reduction Ratio (Final)	3:1

TABLE XI: Summary of Chain-Sprocket Transmission Forces

B. Tractive Force and Acceleration Performance

Parameter	Value
Mass of Go-Kart (with driver)	180 kg
Available Power at Wheels (85% efficiency)	7606.14 W
Maximum Tractive Force (No Slip Condition)	635.69 N
Maximum Velocity	28.55 m/s (102.78 km/h)
Acceleration at 5 m/s	8.26 m/s ²

TABLE XII: Tractive Effort and Performance Summary

C. Dynamic Loading Analysis

- Maximum Transient Chain Tension: 436.84 N (15% overshoot)
- Selected Chain Rating: 520 chain, safe for up to 1000 N load
- All components are within safe operating limits under dynamic conditions.

D. Conclusion

The results validate that the selected transmission system is:

- Capable of delivering high initial acceleration and efficient top-end speed.
- Structurally safe under both static and transient loading.
- Suitable for track-based performance and endurance racing.

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